touch the number 2π on the number line when the circle rolls to one complete revolution to the right.

(iii) If the unit of angle measure is not specified, then the angle is understood to be in radians.

(iv) Consider a sector of a circle with radius r. If θ is the central angle of the sector, then

$$\text{Area of the sector} = \begin{cases} \left(\frac{\pi r^2}{360^\circ}\right)\theta & \text{in degree measure} \\ \left(\frac{\pi r^2}{2\pi}\right)\theta = \frac{r^2\theta}{2} & \text{in radian measure} \end{cases}$$

Clearly, the calculation in radian measure is much easier to work with.

(v) The values of π and $\frac{22}{7}$ correct to four decimal places are 3.1416 and 3.1429 respectively. Thus, π and $\frac{22}{7}$ are approximately equal correct upto two decimal places. Hence, $\pi \approx \frac{22}{7}$.

(vi)

$$1 \text{ radian} \approx 57^\circ 17' 45'' \text{ and } 1^\circ \approx 0.017453 \text{ radian}$$
$$1' = \left(\frac{\pi}{180 \times 60}\right) \text{radian} \approx 0.000291 \text{ radian}.$$
$$1'' = \left(\frac{\pi}{180 \times 60 \times 60}\right) \text{radian} \approx 0.000005 \text{ radian}.$$

(vii) The radian measures and the corresponding degree measures for some known angles are given in the following table

Radians	0	1	0.017453	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
Degrees	0°	$57^\circ 17' 45''$	1°	30°	45°	60°	90°	180°	270°	360°

(viii) $\sin 90^\circ = 1$ but $\sin 90 \neq 1$ (in radian measure).

Example 3.4 Convert (i) 18° to radians (ii)-108° to radians.

Solution:

Now, $180^\circ = \pi$ radians gives $1^\circ = \frac{\pi}{180}$ radians

(i) $18^\circ = \frac{\pi}{180} \times 18$ radians $= \frac{\pi}{10}$ radians

(ii) $-108^\circ = \frac{\pi}{180} \times (-108)$ radians $= -\frac{3\pi}{5}$ radians.

Example 3.5 Convert (i) $\frac{\pi}{5}$ radians to degrees (ii) 6 radians to degrees.

Solution:

We know that π radians $= 180^\circ$ and thus,

(i) $\frac{\pi}{5}$ radians $= \frac{180^\circ}{5} = 36^\circ$

(ii) 6 radians $= \left(\frac{180}{\pi} \times 6\right)^\circ \approx \left(\frac{7 \times 180}{22} \times 6\right)^\circ = \left(343\frac{7}{11}\right)^\circ$.